

Determining the bonding–degradation trade-off at heterointerfaces for increased efficiency and stability of perovskite solar cells

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The heterointerfaces between perovskite and charge-transporting layers pose a major limitation to the durability of perovskite solar cells (PSCs), largely due to complex and conflicting chemical and mechanical interactions. Here we introduce an effective debonding technique to thoroughly analyse heterointerface behaviour during both crystal growth and ageing phases of PSCs. Our analysis reveals a strong correlation between interface bonding (fracture energy ranging from $\sim 2.49 \text{ J m}^{-2}$ to $\sim 0.38 \text{ J m}^{-2}$), proton transfer interactions and degradation, highlighting a critical trade-off between mechanical and chemical stability in PSCs. To address these stability challenges, we mixed Me-4PACz and DCZ-4P molecules, which introduced additional phosphonic acid anchoring groups to enhance bonding at both the metal oxide and the perovskite interfaces. With a high efficiency of 25.6%, the devices retained 90% of their initial performance after 1,000 h of testing under ISOS-L-II and ISOS-D-2I standard protocols. Under thermal cycling conditions, our PSCs sustained 95% of their efficiency over 500 cycles, exceeding the IEC 61215 and ISOS-T-3I standards.

Perovskite solar cells (PSCs) have experienced substantial advancements, achieving power conversion efficiency (PCE) exceeding 26% in single-junction cells and 34% in perovskite–silicon tandem cells¹, and are at a pivotal point in the scale-up from laboratory to factory. Metal oxides, such as nickel oxide (NiO_x), tin oxide (SnO_x) and titanium oxide (TiO_x), have been widely explored and used as charge transport layers (CTLs) in PSCs^{2–9}. However, harmful chemical reactions at the heterointerfaces between CTLs and the perovskite absorber, leading to non-radiative recombination and the formation of initial degradation sites^{10–12}, affect both the initial PCE and the long-term chemical

stability of PSCs. There is limited understanding of the interactions and degradation mechanisms at these interfaces. Unlike the perovskite–organic interface that can be easily separated, enabling thorough investigation of buried interfaces¹³, the specific interactions at perovskite–metal-oxide (Pero– MeO_x) heterointerfaces, particularly during different ageing conditions, remain a subject of ongoing research and discussion^{14–17}. Such highly reactive heterointerfaces often exhibit notable resistance to debonding. However, a comprehensive understanding of the critical relationship between mechanical bonding and chemical degradation at the Pero– MeO_x interfaces remains lacking.

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This underscores the importance of developing methods that can reveal perovskite heterointerfaces and the mechanisms throughout the entire life cycle of solar cells, from initial fabrication to ageing.

Meanwhile, along with the progress of organics such as self-assembled monolayers (SAMs), now the bilayer structure of SAM-metal-oxide is seen as a promising industrial CTL architecture for achieving high-performance PSCs owing to its excellent perovskite passivation effects and hole transport properties¹⁸. On the one hand, although SAMs are expected to work as interface contact barriers blocking chemical reactions between perovskite and metal oxide, the interface degradation cannot be avoided in the long term. On the other hand, interfacial mechanical delamination has emerged as another challenge for such Pero-CTL structures during thermal stability tests owing to the weak interface bonding (Supplementary Fig. 1) and mismatch of the coefficient of thermal expansion^{19–21}. Achieving enduring reliability under the wide temperature fluctuations experienced by solar panels in real-world settings is a double challenge in the efforts to improve the chemical and mechanical stability of PSCs^{22,23}. This longevity performance can be well measured by thermal cycling testing according to International Electrotechnical Commission (IEC) and International Summit on Organic Photovoltaic Stability (ISOS) protocols^{22,24}. Therefore, striking a balance between robust mechanical bonding and preventing degradation reactions at the Pero-CTL interface is especially important. However, achieving standardized thermal cycling stability remains a challenge in the development of efficient PSCs²⁵. Furthermore, the IEC 61215 standard (200 cycles, dwell time 10 min), which is now adopted in laboratory, is still far from practical for industrial stability testing (more than 500 cycles, longer dwell time)²⁴.

In this study, we developed a debonding technique to uncover strongly bonded/highly reactive and weakly bonded/lowly reactive heterointerfaces in PSCs. This approach allowed us to perform an in-depth analysis of the interactions at various stages, including the initial crystal growth phase and the long-term ageing phase. Our findings show a correlation between interface bonding, proton transfer interactions and degradation during perovskite crystal growth. We also found that heat ageing amplified proton transfer reactions, whereas electron-hole transfer reactions were more prominent during light ageing. To overcome the paradox between mechanical bonding and chemical degradation, we mixed [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz) and 9H,9'-H-[3,3'-bicarbazole]-9,9'-diylbis(butane-4,1-diyl)diphosphonic acid (DCZ-4P) SAM molecules, which introduced additional phosphonic acid anchoring groups to enhance bonding at both the metal oxide and perovskite interfaces. This approach strengthened heterointerface bonding while minimizing adverse chemical reactions. Our PSCs not only achieved a maximum efficiency of 25.6% (certified 25.4%) but also demonstrated improved stability, maintaining 90% efficiency after 1,000 h of operational and thermal ageing tests (ISOS-L-II and ISOS-D-2I). The target PSCs kept 95% of their initial PCE over 500 cycles of more rigorous thermal cycling evaluation (dwell time 30 min, beyond the IEC 61215 standard; ISOS-T-3I).

Interface mechanical performance

We developed a non-destructive, fabrication-compatible and universally effective technique to uncover interactions at the Pero-MeO_x interfaces. Our approach began with bonding two half-cells, each comprising a perovskite layer (Cs_{0.1}FA_{0.9}PbI₃) on various metal-oxide substrates, under vacuum condition. This was followed by a debonding process to separate the Pero-MeO_x interfaces (as illustrated in Fig. 1a). The high crystallinity within the bonded perovskite layers allowed the initially distinct films to merge seamlessly, making robust heterogeneous Pero-MeO_x interfaces prone to debonding. Cross-section scanning electron microscopy (SEM) imaging showed intact, uniformly crystallized bulk perovskite layers formed under vacuum-bonding conditions, contrasting with perovskite laminated under atmospheric conditions

that displayed a clear boundary, indicative of weaker Pero-Pero bonding (Fig. 1b). The use of a high vacuum during the bonding process mitigated the adverse effects of moisture and oxygen, previously identified as key negative factors in PSCs²⁶. X-ray diffraction (XRD) patterns indicated lower PbI₂ peaks in vacuum-bonded perovskite compared with traditionally laminated perovskite (Fig. 1c)²⁷. Photoluminescence (PL) spectroscopy further supported this, showing lower PL intensity and energy band narrowing due to defects in traditionally laminated perovskite (Supplementary Fig. 2). Consequently, the subsequent debonding at the Pero-MeO_x interfaces is primarily mechanical. Notably, this bonding process closely mimics the perovskite deposition step in our solar cell fabrication process, enabling effective monitoring of the Pero-MeO_x interfaces before PSC operation⁵.

By utilizing mechanical double-cantilever beam (DCB) testing method while debonding the MeO_x-Pero-MeO_x stack (as shown in Supplementary Fig. 3 and Methods), the interface bonding performance of various common Pero-MeO_x configurations can be demonstrated by interface fracture energy (Fig. 1d)^{2–9}. The value of fracture energy can be measured from the testing load-displacement curve (Supplementary Fig. 4 for ZnO_x-Pero-ZnO_x). Notably, we discovered a ranking of interface bonding from Pero-ZnO_x with the highest average fracture energy (−2.49 J m^{−2}) to Pero-TiO_x with the lowest (−0.38 J m^{−2}). Higher fracture energy will be beneficial to avoid delamination at the heterointerface (Supplementary Fig. 5). Within the scope of our study, the effects of surface roughness and nanocrystal size are negligible (Supplementary Fig. 6). The difference in specimen preparation and condition can cause different measured values for interface fracture energy^{28,29}, which also indicates the importance of further process standardization. Benefitting from strong lattice bonding within perovskite bulk, our developed debonding technique is a promising alternative adding to the epoxy-peeling method³⁰. Based on this approach, we carry out further systematic studies for heterointerfaces.

Characterization and analysis of heterointerface defects

After debonding, we first conducted different imaging characterizations to visualize the buried perovskite in various Pero-MeO_x interfaces and summed up their parameters extracted from the images. SEM and atomic force microscopy (AFM) were utilized to show the morphology of various revealed perovskite (Fig. 2a,b). Voids of varying sizes were found in nearly all of our images of the Pero-MeO_x interfaces, and these voids were shown to be defective regions that affect PSC efficiency and stability¹³, but the area ratio of the voids varied from interface to interface. The perovskite revealed on ZnO_x exhibits an unusual morphology, unevenly covered with hexagonal impurities that were proven to be PbI₂ by the elemental ratios obtained by energy-dispersive X-ray spectroscopy (Supplementary Fig. 7). Thus, strong interactions occur at the Pero-ZnO_x interface, leading to the formation of a substantial amount of lead iodide during the perovskite crystallization process. The above two regions (voids and impurities) in SEM images are considered microscopy-visible non-perovskite areas on the revealed perovskite, which are identified over a 100 μm² area by image processing to extract their proportion of the overall area (shown in the histogram in Fig. 2e). For AFM, surface roughness values were extracted to verify the trend that the severity of disruption for the bottom-interface perovskite morphology, due to interfacial interactions during crystal growth, decreases from Pero-ZnO_x to Pero-TiO_x (Fig. 2b,e).

To study the effect of Pero-MeO_x interfacial interactions on the optical and electrical properties of perovskite films, we further conducted confocal PL mapping and kelvin probe force microscopy (KPFM) measurements for same series of revealed perovskite. As shown in Fig. 2c,e, revealed perovskite films showed higher overall PL intensity and increased uniformity from Pero-ZnO_x to Pero-TiO_x. Dark areas, which indicate low PL emission intensity, were observed to be the most prevalent on the perovskite in the Pero-ZnO_x interface. These were

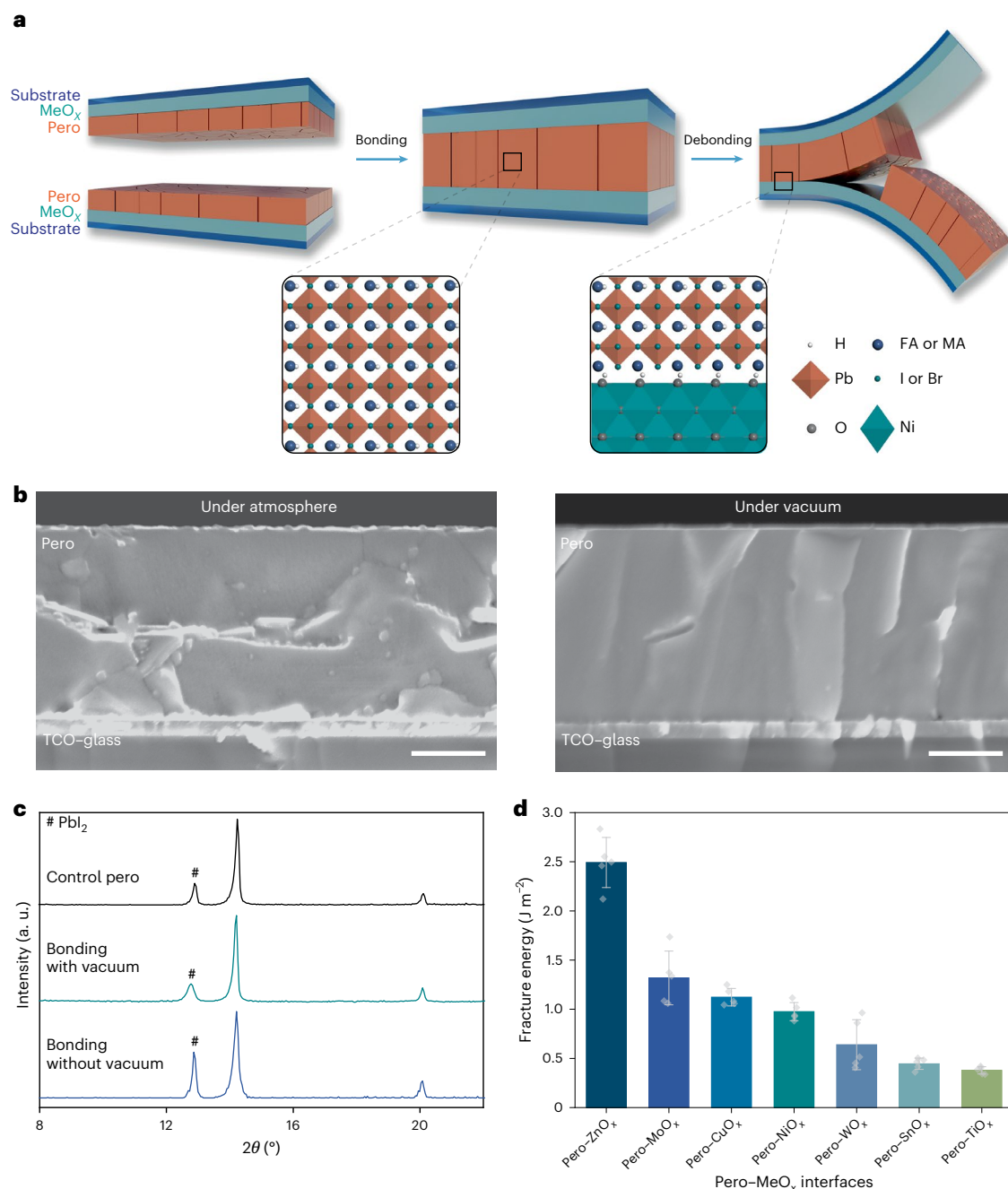


Fig. 1 | Methodology of debonding technique and interface bonding.
a, A diagram of the bonding and debonding process in the technique (NiO_x is set as an example in the zoom-in structure. FA, formamidinium; MA, methylamine).
b, Cross-section SEM images of atmosphere-laminated (left) and vacuum-bonded (right) perovskite films. Scale bars, 500 nm. TCO, transparent

conducting oxide. **c**, An XRD pattern comparison for atmosphere-laminated and vacuum-bonded perovskite films. **d**, DCB-tested fracture energy for various Pero-MeO_x interfaces, each based on five statistical samples (data are presented as mean values ± s.d.).

attributed to the initial degradation sites of local defects and heterogeneity at the interface that had arisen during crystal growth, which also corresponded to the morphological regions discussed above. KPFM surface potential distributions are measured for all samples (Fig. 2d). Compared with that of control perovskite sample coated on glass (Supplementary Fig. 8), we noted that the surface potential of revealed perovskite showed negative or positive shifts for various Pero-MeO_x interfaces (Supplementary Fig. 9). The potential variations, which are influenced by various interface defects³¹, are found to get smaller for interface perovskite from Pero-ZnO_x to Pero-TiO_x (Fig. 2e). The uniform distribution of interface potential facilitates effective carrier

extraction and stable device operation³². We deduce that interfacial degradation of perovskite already occurred in the device fabrication process and led to the formation of initial interfacial defects. This degradation can be attributed to the interactions at various Pero-MeO_x interfaces, based on the striking consistency between the degradation trends in Fig. 2e and the interface bonding trend in Fig. 1d.

Heterointerface bonding-degradation mechanism

To investigate the mechanism of Pero-MeO_x interfacial interactions, we further conducted X-ray photoelectron spectroscopy (XPS) to study

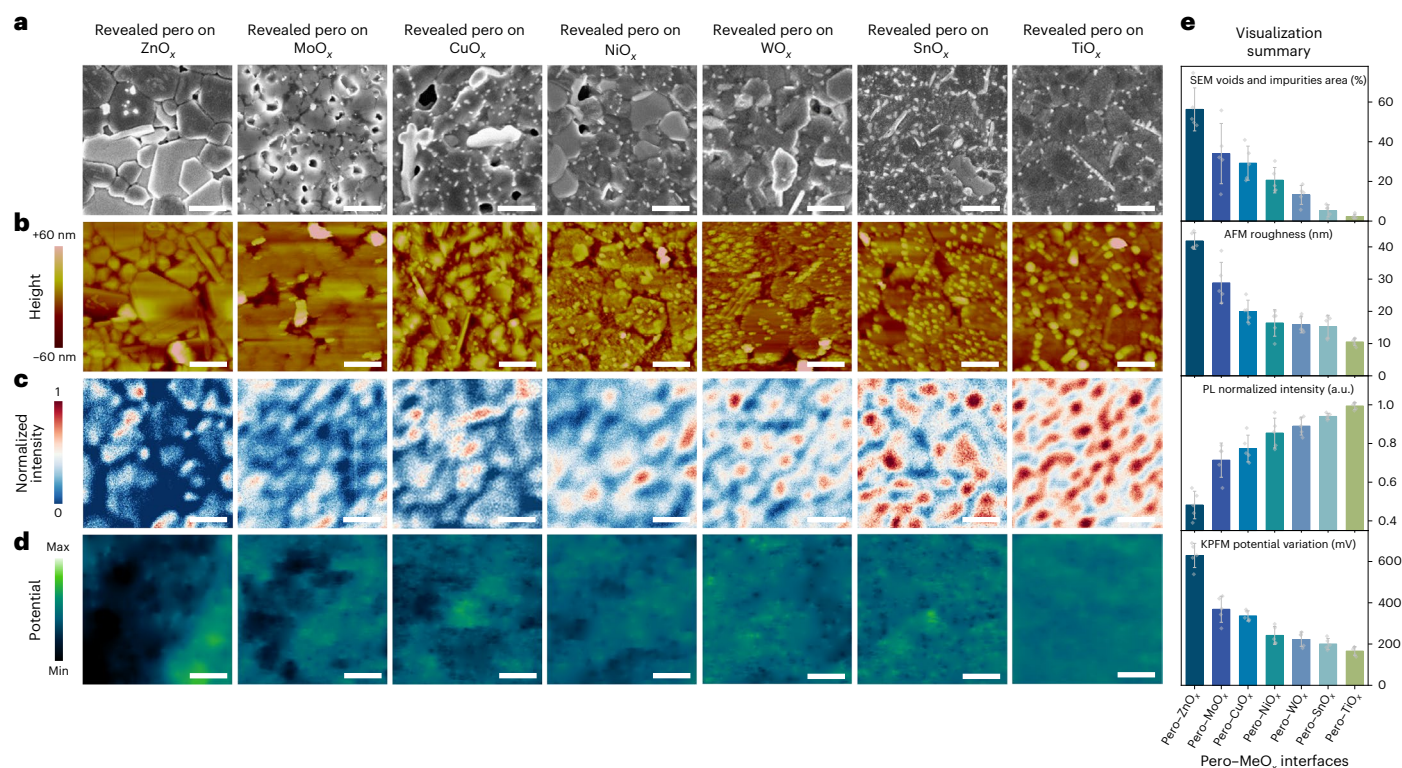


Fig. 2 | Visualization characterizations of perovskite after debonding. **a–d**, Various revealed perovskites on ZnO_x, MoO_x, CuO_x, NiO_x, WO_x, SnO_x and TiO_x visualized by SEM (scale bars, 1 μm) (**a**), AFM (scale bars, 1 μm) (**b**), confocal PL mapping (scale bars, 1 μm) (**c**) and KPFM (scale bars, 500 nm) (**d**).

e, A visualization summary for the revealed perovskites in various Pero–MeO_x interfaces by extracting the respective characteristic parameters of different visualization methods, each based on five statistical samples (data are presented as mean values ± s.d.).

the electronic structures of major elements in revealed perovskite and metal oxides. For the revealed perovskite in the typical Pero–ZnO_x, the binding energies of Pb 4f 7/2 and I 3d 5/2 positively shift -0.20 eV synchronously compared with the control sample coated on glass (Supplementary Fig. 10a,b), which represents the transformation from perovskite to PbI₂ due to loss of A⁺I⁻ (where A⁺ is generally the cations) coordination³³. Besides, in the N 1s spectrum of the revealed perovskite sample (Supplementary Fig. 10c), an additional peak marking the deprotonated FA (CH₃N₂) appears at a lower energy position¹⁴, while the control sample shows only one characteristic peak at -400.7 eV representing the N atoms coordinated to C and H in the formamidinium cation (CH₅N₂⁺, FA⁺). From the O 1s and Zn 2p spectra of revealed ZnO_x (Supplementary Fig. 10d,e), we observed that oxygen vacancy in ZnO_x increased after contacting with perovskite, and the Zn 2p peak shift of +0.7 eV can be attributed to the transformation of Zn–O bonds into Zn–OH and Zn–I bonds. From Fig. 3a and Supplementary Figs. 10–17, we can further infer that all the Pero–MeO_x interfaces were undergoing the same reaction processes with different degrees, including FA deprotonation and PbI₂ production (supported by the Pb/I ratio extracted from XPS spectra in Supplementary Fig. 18) for perovskite as well as oxygen vacancy formation for metal oxides at the heterointerface. The pre-stage interfacial reaction mechanism of perovskite on metal oxides can be considered as the following process: the metal oxide interacts with protons (H⁺) and halogen ions (here I⁻) in the perovskite, leading to oxygen vacancy formation, while the FA⁺ is deprotonated and the remaining Pb²⁺ and I⁻ transform to lead iodide.

In Fig. 3a, the extent of perovskite deprotonation decreased from Pero–ZnO_x to Pero–TiO_x, confirmed by bulk XRD and Fourier transform infrared spectra of various Pero–MeO_x films (Supplementary Figs. 19 and 20). Notably, Fig. 3b shows a strong correlation between the intensity of interfacial proton transfer interaction and interface bonding of various heterointerfaces. Accordingly, we propose that the interface

bonding can originate from the adsorption of protons in the perovskite by metal oxides. The strong adsorption capacity strengthens the interface bonding but simultaneously increases interfacial proton transfer, which can deteriorate the Pero–MeO_x interface stability. We also show the general applicability of this mechanism to perovskite, including bromine-containing (40%) wide bandgap perovskite (Br-Perov) (Supplementary Fig. 21). X-ray absorption fine structure (XAFS) analysis was further conducted to validate our hypothesis by probing the structural properties of the revealed ZnO_x. As shown in the X-ray absorption near-edge structure (XANES) results (Fig. 3c), the control ZnO_x (before contacting with perovskite) is nearly the same as standard ZnO, with typical energy peak positions at 9,671.2 eV, 9,682.5 eV, 9,716.6 eV and 9,740.6 eV. However, the revealed ZnO_x (after interaction with perovskite during crystal growth) showed obviously different features compared with that of the control one, and was close to Zn(OH)₂. The approximate transition from zinc oxide to zinc hydroxide provides direct evidence for the adsorption of protons from perovskite cations by metal oxides. It was reported that proton adsorption capacity depends mainly on the inherent isoelectric point of different metal oxide surfaces³⁴ and can be tuned by oxygen vacancy density and component^{35,36}.

In addition to monitoring interfacial interactions during perovskite crystal growth, our debonding technique allows us to examine the revealed interface after it has been subjected to various ageing conditions. First, we investigated the effect of heat ageing (100 h) on the heterointerface (Fig. 3d). After revealing the interfaces, we observed that the high-temperature condition (85 °C) accelerated the deprotonation reaction at the perovskite interface, resulting in increased PbI₂ formation. This is a common occurrence, even in Pero–TiO_x systems, in which the initial reaction is typically milder. This is due to high temperatures reducing the activation energy for proton transfer³⁷, which obviously will cause more serious device degradation.

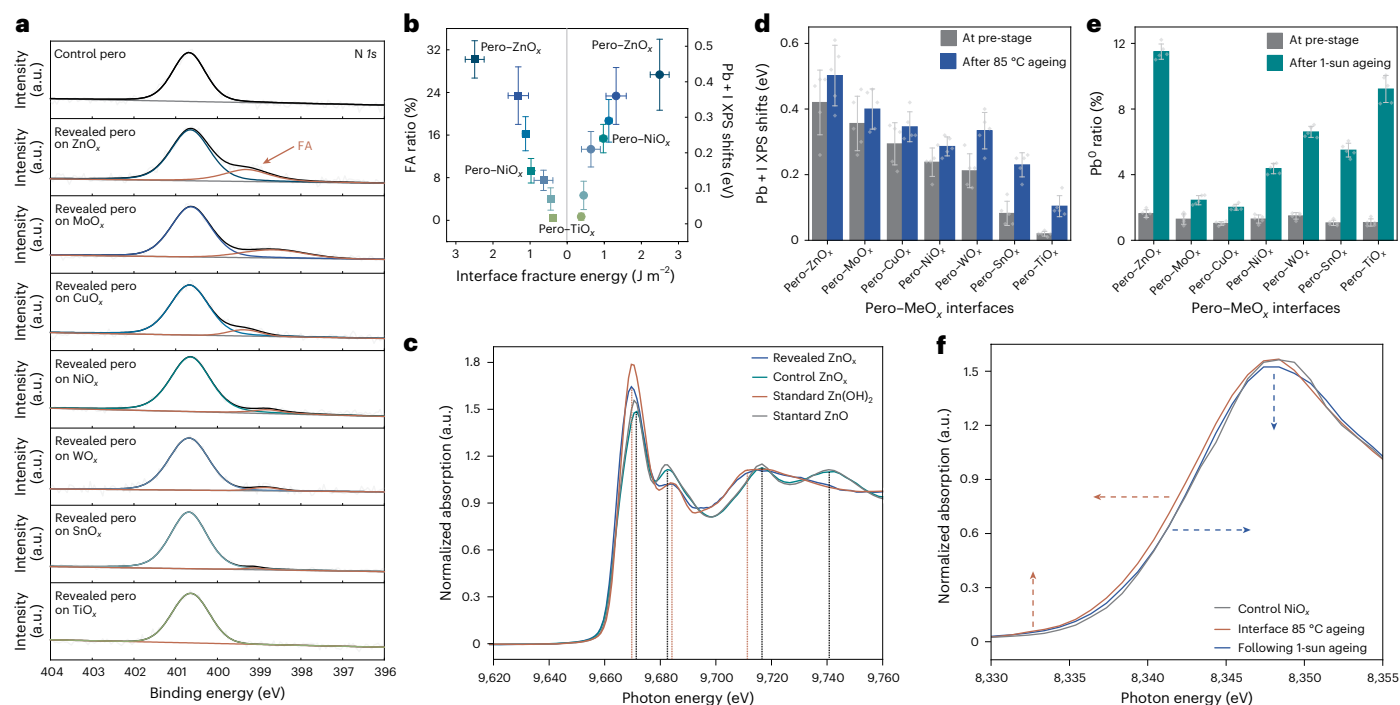


Fig. 3 | Interface chemical reaction characteristics. **a**, An XPS N1s spectra comparison of control perovskite film and various revealed perovskite films on ZnO_x, MoO_x, CuO_x, NiO_x, WO_x, SnO_x and TiO_x (fitting curves by XPSPEAK, linear background). **b**, The correlation between interface fracture energy and interface fracture energy, each based on five statistical samples (data are presented as mean values ± s.d.). **c**, Zn K-edge XANES spectra (9,620–9,760 eV) of ZnO_x in the Pero-ZnO_x interface at the pre-stage of PSCs (the vertical lines indicate characteristic peaks of standard Zn(OH)₂ and ZnO, respectively).

d, The sum of the Pb 4f_{7/2} and 13d_{5/2} core level shift extracted from XPS spectra for various revealed perovskites, each based on five statistical samples (data are presented as mean values ± s.d.). **e**, The metallic Pb ratio extracted from XPS spectra of Pb 4f, each based on five statistical samples (data are presented as mean values ± s.d.). **f**, In situ Ni K-edge XANES spectra (8,330–8,355 eV) of NiO_x in the Pero-NiO_x interface subjected to continuous 85 °C and following 1-sun ageing (the arrows indicate the absorption edge shifts and peak height changes, respectively).

Second, we examined the interfaces that had undergone 100 h of 1-sun light ageing. The XPS peak shifts of most revealed perovskite under light ageing are similar to those after crystal growth (Supplementary Fig. 22). However, more metallic lead is observed at all Pero-MeO_x interfaces after light ageing (Fig. 3e), for which the different degradation pathway may involve a redox reaction due to interface contact-induced electron-hole transfer¹⁵, including photodecomposition of lead iodide and iodine oxidation^{38–40}. It is also noteworthy that the proportion of metallic Pb produced at different Pero-MeO_x interfaces varies. It has been observed that the electron transport interfaces are more prone to redox interactions.

To clearly elucidate the interfacial interaction mechanisms under different ageing conditions, we examined the structural geometry variation of NiO_x during continuous ageing at the Pero-NiO_x interface using in situ XAFS (Fig. 3f)⁴¹. First, under heat ageing, the absorption edge (8,339–8,346 eV) of NiO_x showed a redshift (–0.5 eV) and the pre-edge peak (8,330–8,339 eV) increased. This indicates a notable decrease in the Ni valence states attributed to the enhanced H⁺ adsorption, which causes an increase in the 3d orbital occupancy and, thus, possible distortion of NiO₆ octahedron in NiO rock-salt structure⁴². The light ageing afterwards resulted in a relative blueshift (–0.4 eV) of the absorption edge, along with a height decrease of the peak at 8,348 eV. This suggested that holes excited at the Pero-NiO_x interface by light are transferred to NiO_x, resulting in a recovery of the Ni valence states. It is noteworthy that the characteristic increase in the pre-edge peak due to heat ageing is still retained in light ageing, suggesting that the proton transfer interaction is irreversible to some extent.

We elucidate the correlation between interface bonding, reactions and degradation at the Pero-MeO_x heterointerfaces, whereby the adsorption of protons in perovskites by metal oxides mechanically

manifests as bonding at the interfaces (as in Fig. 1), chemically triggers interfacial proton transfer reactions and leads to the formation of interfacial defects and degradation sites (as in Figs. 2 and 3). Under these circumstances, a paradox arises at the Pero-MeO_x interface: if gentle interfacial reactions and minimal interfacial defects are desired, the result is often weak interface bonding, which, in turn, increases the susceptibility to mechanical delamination during operation.

Optimization strategy for PSCs

The Pero-NiO_x interface is one of the most crucial interfaces in state-of-the-art p-type/intrinsic/n-type (p-i-n) PSCs^{41,43}. By using this debonding method, we proved that commonly used Me-4PACz introduced into the Pero-NiO_x interface can work as a barrier interlayer for preventing interfacial reactions (with less FA deprotonation, as shown in Fig. 4a and Supplementary Fig. 23). However, Me-4PACz also induces a delamination problem due to its weak bonding with perovskite (Supplementary Fig. 1). The carbazole group at the top of Me-4PACz-NiO_x is thought to effectively block proton transfer at the interface, which also resulted in weaker interface bonding (fracture energy of –0.41 J m⁻²) than NiO_x (–0.98 J m⁻²) and, thus, causes the interface vulnerability. To further manage the bonding-degradation trade-off, we introduced a molecule, DCZ-4P (Supplementary Fig. 24), which was mixed in a 1:1 molar ratio with Me-4PACz to form a mixed SAM interlayer, referred to as D4P. As illustrated in Fig. 4b, the mixed SAMs of Me-4PACz, containing a single phosphonic acid group, and DCZ-4P, featuring double phosphonic acid groups, provided additional phosphonic acid groups for interaction with the perovskites, thereby enhancing interfacial bonding and stability. The increased concentration of phosphonic acid groups at the Pero-CTL interface in D4P was confirmed by time-of-flight secondary ion mass spectrometry (ToF-SIMS; Fig. 4c). The inhibited

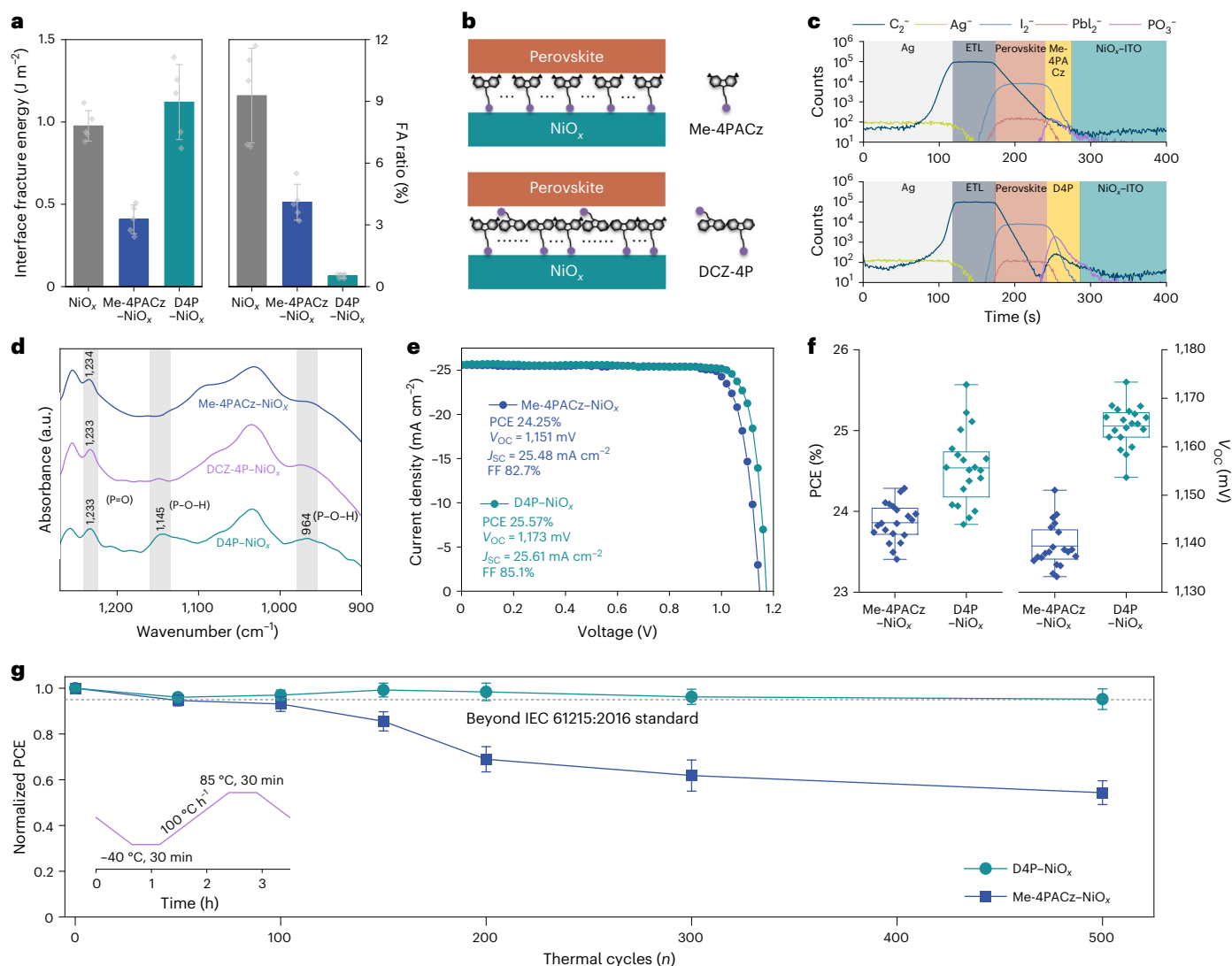


Fig. 4 | Heterointerface characteristics and PSC device performance.

a, A comparison of interface fracture energy and deprotonation reaction intensity for Pero-NiO_x, Pero-Me-4PACz-NiO_x and Pero-D4P-NiO_x, based on five statistical samples (centres: mean; error bars: s.d.). **b**, An illustration of Pero-Me-4PACz-NiO_x (up) and Pero-D4P-NiO_x (down) interface architecture. **c**, ToF-SIMS results for PSCs based on Me-4PACz-NiO_x and D4P-NiO_x architecture. **d**, IRRAS spectra of Me-4PACz-NiO_x, DCZ-4P-NiO_x and D4P-NiO_x. **e, f**, Comparisons for

PSCs based on Me-4PACz-NiO_x and D4P-NiO_x architecture of *J*-*V* curves (**e**) and statistics of PCE (left) and *V*_{OC} (right) (based on 20 statistical samples, whiskers: maxima and minima; bounds of box: 25th and 75th percentile; centre: mean) (**f**). **g**, Normalized PCE tracking under thermal cycling for respective PSCs (based on four statistical samples, data are presented as mean values $\pm$ s.d.). Inset: temperature-time curve of thermal cycling stability test (beyond IEC 61215 standard; ISOS-T-3I).

migration of iodine (I₂⁻) from perovskite to the bottom layer implied in ToF-SIMS is consistent with XPS results showing that D4P is a more effective barrier to interfacial reactions (with nearly no FA deprotonation as shown in Supplementary Fig. 25), which can be contributed by the parallel carbazole groups⁴⁴. Notably, we observed that, with a better reaction-blocking effect, D4P exhibits stronger interface bonding (fracture energy of $\sim 1.12 \text{ J m}^{-2}$) with perovskite than NiO_x (Fig. 4a). The stronger bonding is proposed to originate from the top-side phosphonic acid group interacting with perovskite⁴³. After debonding, both the revealed surfaces of the Pero-D4P-NiO_x interface showed P signals, indicating that D4P bonded on both sides (Supplementary Fig. 26). The 1:1 molar ratio of Me-4PACz and DCZ-4P demonstrated the best performance (Supplementary Fig. 27), confirming that D4P effectively exposed high concentrations of phosphonic acid groups at the top surface. In contrast, pure DCZ-4P did not show notable interface optimization effects, which may be attributed to its face-on orientation and sparse alignment due to the larger molecule size (Supplementary Fig. 28)⁴⁵. By comparing the infrared reflection absorption

spectroscopy (IRRAS) spectra (Fig. 4d), much larger intensity of peaks at 1,145 cm⁻¹ and 964 cm⁻¹ was observed for D4P-NiO_x. Both bands are attributed to the antisymmetric and symmetric stretching vibrations of P-O-H (ref. 45), which bridges the Pero-CTL interface with strong bonding. Moreover, the water contact angle (Supplementary Fig. 29) of D4P-NiO_x was shown to be 40.0°, much less than that of Me-4PACz-NiO_x (65.8°) and DCZ-4P-NiO_x (72.1°), suggesting that there are more hydrophilic phosphonic acid groups on D4P-NiO_x. With the enhanced interlayer structure, the perovskite revealed at the Pero-CTL interface also showed much more uniform SEM morphology and PL imaging with fewer defects (Supplementary Fig. 30 compared with NiO_x in Fig. 2).

The current density-voltage (*J*-*V*) curves of the champion p-i-n-based Me-4PACz-NiO_x PSCs and D4P-NiO_x PSCs from the same batch were recorded (Fig. 4e), showing that D4P-NiO_x architecture can benefit PCE, increasing to 25.6%. External quantum efficiency measurement (Supplementary Fig. 31) showed that the optical performance of the two architectures is identical. The PCE of the champion D4P-NiO_x PSCs was certified as 25.4% (Supplementary Fig. 32). The better PCE

achieved by D4P–NiO_x PSCs is mainly due to the higher open-circuit voltage (V_{oc}) (Fig. 4f and Supplementary Fig. 33), benefitting from minimized interface recombination. XRD patterns of Pero–Me-4PACz–NiO_x and Pero–D4P–NiO_x did not show differences in perovskite bulk phase, but PL spectra results obtained from the bottom layer indicated that Pero–CTL interface defects are greatly reduced by introducing D4P (Supplementary Fig. 34). We calculated the quasi-Fermi level splitting values or implied V_{oc} against the illumination intensity by intensity-dependent steady-state PL measurements and then constructed pseudo- j - V for both Me-4PACz–NiO_x and D4P–NiO_x stacks in comparison (Supplementary Fig. 35)⁴⁶. It is observed that D4P–NiO_x effectively minimized voltage loss.

To verify that the D4P–NiO_x architecture is stable in PSCs, we applied several standard stability tests²². By comparing the long-term operation stability under the 1-sun maximum power point tracking condition (25 °C, relative humidity of 85%; ISOS-L-1I), it can be found that T_{90} (the operation time after which the solar cell retains 90% of its initial PCE) of D4P–NiO_x PSCs is more than 1,000 h (Supplementary Fig. 36a). However, the normalized PCE of Me-4PACz–NiO_x PSCs dropped to below 80% after 660 h. Furthermore, after ageing encapsulated solar cells in an 85 °C environment (ISOS-D-2I) and tracking their PCE for over 1,000 h, it was found that D4P–NiO_x PSCs can maintain 91.3% of their initial PCE compared with 73.2% for Me-4PACz–NiO_x PSCs (Supplementary Fig. 36b). The enhanced thermal and light stability can be attributed primarily to the improved bonding and reaction-blocking effect provided by D4P. In the thermal cycling test (Fig. 4g), a challenging 30 min dwell condition (–40 °C/85 °C, exceeding IEC 61215 and ISOS-T-3I standards) was employed to rigorously assess both the chemical and mechanical stability of the PSCs. After 500 thermal cycles, the D4P–NiO_x PSCs retained 95% of their initial PCE, compared with just 54% for Me-4PACz–NiO_x PSCs. This demonstrates that the higher fracture energy at the Pero–D4P–NiO_x interface effectively prevents harmful mechanical delamination during thermal cycling (as shown in Supplementary Fig. 37).

Conclusion

By developing an effective debonding technique, we uncovered the mechanisms of PSC heterointerface interactions across various stages, from initial crystal growth to ageing. Our findings highlight the paradoxical relationship between mechanical bonding and chemical degradation at the heterointerfaces. We demonstrate that, for Pero–MeO_x interfaces, mechanically strong interfacial bonding is often accompanied by intense chemical reactions. Proton transfer reactions play a pivotal role, leading to the formation of interfacial defects and degradation sites. To resolve this bonding–degradation trade-off, the rational design of SAMs, which block harmful interfacial reactions while strengthening interface bonding, offers a promising strategy for optimizing perovskite heterointerfaces. This approach not only enhances the initial performance of PSCs but also notably improves their long-term stability under rigorous conditions.

Methods

Materials and preparation

Caesium iodide (CsI, 99.99%) lead chloride (PbCl₂), dimethylformamide (DMF, anhydrous) and 1-methyl-2-pyrrolidinone (NMP, anhydrous) were purchased from Sigma-Aldrich. Formamidinium iodide (FAI) and ethane-1,2-diammonium iodide were purchased from Greatcell Solar Materials; anhydrous lead iodide (PbI₂) and Me-4PACz were purchased from Tokyo Chemical Industry. Silver was purchased from Alfa Aesar, and C₆₀ and bathocuproine (BCP) were purchased from Lumtec. ZnO_x, MoO_x, NiO_x, WO_x, SnO_x and TiO_x nanoparticle ink solutions were purchased directly from Sigma-Aldrich. For CuO_x, copper(II) acetylacetonate (Cu(acac)₂, purity >99.9%) were purchased from Sigma-Aldrich as the precursor for solution-processing preparation. DCZ-4P is self-synthesized.

Technique procedure

For half-cells, the tin-doped indium oxide (ITO) glass substrates were wiped with soap water, then ultrasonically washed with deionized water, acetone and isopropanol for 30 min each. After ultraviolet ozone treatment for 15 min, the substrates were spin-coated with as-purchased MeO_x nanoparticle ink, followed by annealing in air at 100 °C for 10 min for MoO_x, NiO_x and WO_x layers, and at 150 °C for 30 min for ZnO_x, SnO_x and TiO_x layers, respectively. The CuO_x layer was prepared as reported in a previous publication². Cu(acac)₂ in 1,2-dichlorobenzene solution (1.5 mg ml^{–1}) was spin-coated and then annealed at 100 °C for 20 min in air. After washing with methanol, the film was transferred into ultraviolet ozone for treatment for 15 min. The 1.8 M perovskite (Cs_{0.1}FA_{0.9}PbI₃) stock solution contains 52 mg CsI, 309.6 mg FAI, 922 mg PbI₂ and 55.6 mg PbCl₂ that were dissolved in 1.11 ml mixed solvent of DMF:NMP (v:v = 960:150). The 1.8 M perovskite precursor solution was shaken overnight to fully dissolve it and was then used for preparing perovskite films. For the spin-coating process, the substrate was spun at 5,000 r.p.m. for 40 s with an acceleration of 5,000 r.p.m. s^{–1} and N₂ gas was blown on top of the spinning substrates after 20 s. The perovskite films were pre-annealed at 70 °C for 2 min in a N₂ glovebox. So far, the perovskite deposition process is the same as that in ‘Device fabrication’ section.

For the debonding technique, the two as-prepared glass–ITO–MeO_x–Pero half-cells were immediately transferred into a wafer bonding tool (EVG 501). In a high-vacuum environment, the two half-cells were piled perovskite to perovskite and annealed under –1,000 N at 150 °C for 10 min. Afterwards, the samples were taken out and found to be tightly bonded. The half-cells were debonded by applying tensile force, after which one can observe on one half-cell a dark area representing the bonded perovskite of which the surface is the revealed perovskite at the other Pero–MeO_x interface, and a transparent area representing the revealed MeO_x on the local Pero–MeO_x interface. The revealed perovskite and MeO_x were then used for characterizations. To reveal the heterointerfaces after ageing, we put the bonded cell on a 85 °C hot plate or under 1-sun for 100 h and then processed the debonding for the heterointerfaces of interest.

For DCB testing, the ‘sandwich’ DCB specimens were fabricated as described above. Aluminium tabs were glued to the glass substrates on both sides of the DCB specimen at the cracked ends. The specimens were then carefully loaded in tension under displacement control on an Instron 5500 equipped with a 100 N load cell. The load (P)–displacement (Δ) curve was recorded at all times during the loading–unloading cycle. During loading, crack growth occurs at the critical load P_c where the P – Δ curve exhibits nonlinearity. The crack length a can be estimated with the elastic compliance $d\Delta/dP$ using the relation

$$a = \left(\frac{d\Delta}{dP} \cdot \frac{EBh^3}{8} \right)^{1/3} - 0.64h,$$

where B and h is the width and half-thickness (0.7 mm) of the DCB specimen, respectively, and E is the elastic modulus (70 GPa) of the glass substrate. The interfacial energy (fracture toughness) G_c can then be calculated with P_c and a via the relation

$$G_c = \frac{12P_c^2 a^2}{B^2 E h^3} \left(1 + 0.64 \frac{h}{a} \right)^2.$$

After reaching P_c , the specimens were unloaded slightly and then loaded again to P_c . The process was repeated until failure. An average value of G_c was then calculated. All DCB testing was carried out in a laboratory air environment (–25 °C and relative humidity of ~45%). The fractured surfaces were subsequently characterized.

Device fabrication

The patterned ITO glass substrates were wiped with soap water, then ultrasonically washed with deionized water, acetone and isopropanol

for 30 min each. After ultraviolet ozone treatment for 15 min, the substrates were then spin-coated with NiO_x (10 mg ml^{-1} in water) nanoparticles, followed by heating at 100 °C for 10 min in the open air. Then, Me-4PACz (3 $\mu\text{mol ml}^{-1}$ in isopropanol) or D4P (Me-4PACz 1.5 $\mu\text{mol ml}^{-1}$ and DCZ-4P 1.5 $\mu\text{mol ml}^{-1}$ together in isopropanol) was spin-coated on the substrates at 4,000 r.p.m. for 30 s and heated at 100 °C for 10 min in the nitrogen filled glovebox. The 1.8 M perovskite ($\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$) stock solution contains 52 mg CsI, 309.6 mg FAI, 922 mg PbI_2 and 55.6 mg PbCl_2 that were dissolved in 1.11 ml mixed solvent of DMF:NMP (v:v = 960:150). The 1.8 M perovskite precursor solution was shaken overnight to fully dissolved it and was then used for preparing perovskite films. For the spin-coating process, the substrate was spun at 5,000 r.p.m. for 40 s with an acceleration of 5,000 r.p.m. s^{-1} , and N_2 gas was blown on top of the spinning substrates after 20 s. The perovskite films were pre-annealed at 70 °C for 2 min in an N_2 glovebox and then annealed at 150 °C for 10 min in air. Afterwards, the samples were transferred back to the glovebox and 1 mg ml^{-1} ethane-1,2-diammonium iodide solution (in isopropanol) was spin-coated onto perovskite film, followed by annealing at 100 °C for 10 min. Lastly, C_{60} (20 nm), BCP (8 nm) and Ag (100 nm) were deposited to complete the device fabrication. The devices were masked with metal aperture mask (0.05 cm^2) during the J - V measurement. A control perovskite sample was fabricated by conducting the same perovskite spin-coating and annealing process directly on the clean glass after ultraviolet ozone treatment for 15 min.

Material characterizations

SEM images were taken with the Regulus SU8200 system (Hitachi) at 3 kV accelerating voltage under secondary electron (SE) mode. XRD was recorded on an X-ray diffractometer (Malvern Panalytical) with $\text{Cu K}\alpha$ radiation ($\lambda = 0.1542$ nm). PL quantum yield was conducted with an LP20-32 radiative efficiency meter (QYB). For suns- V_{oc} measurements, the laser was switched on 15 min before the measurement to allow it to warm up and stabilize. The laser was then set to the desired intensities and performed a PL measurement. For the pseudo- J - V , 90% of Shockley-Queisser (S-Q) current density is used to avoid potential influences for the implied V_{oc} comparison. The DCB test was conducted on the Instron 5500 Testing System (Instron). The samples after bonding were loaded and subjected to tensile testing with a linear strain rate of 1.00 mm min^{-1} , with the requisite applied force recorded until mechanical failure occurred. AFM and KPFM measurements were conducted with the Park NX20 system. AFM measurements were performed with non-contact mode. Sideband KPFM measurements were performed under non-contact mode. The work function was calibrated before every sample with a standard gold reference sample. Also, the signals were optimized in advance by applying bias voltage onto the samples in advance of the measurements. Confocal PL mapping was measured on a Nikon A1 confocal microscope equipped with a 633 nm continuous-wave laser. XPS measurements were performed with a Kratos Axis Ultra XPS system. Samples were stored under high-vacuum conditions overnight to minimize influences from the absorbed H_2O and other particles on the surface. The original data were processed by XPSPEAK41 Software. The operando and ex situ Ni K-edge XAFS and high-resolution XAFS data were collected at the XAFCA beamline of Singapore Synchrotron Light Source under the transmission mode. The storage ring of Singapore Synchrotron Light Source was running at 0.7 GeV with an electron current of 200 mA. Energy calibrations were carried out using standard Ni foil. Athena was used for the energy calibration, background removal and Fourier transform.

Device characterization

J - V measurements of the devices were recorded with a Keithley 2400 source meter under simulated 1-sun air mass (AM) 1.5 G illumination (100 mW cm^{-2}) with an ABET Technologies Sun 2000 solar simulator (with 400–1,100 nm total spectral mismatch of 1.002). The measurements were performed in an N_2 filled glovebox. The illumination

area of the devices was defined by shadow masks of 0.05 cm^2 area. Maximum power point tracking measurements of the encapsulated devices were recorded with an MPP Tracking-4B system (Shenzhen Lancheng Technology) with an LED simulated AM 1.5 G spectrum. The measurements were carried out at ambient conditions at 25 °C with an relative humidity of 85% (ISOS-L-1I). Heat-ageing stability performance was tested by putting encapsulated solar cells in an 85 °C environment chamber and conducting measurements periodically (ISOS-D-2I). Thermal cycling stability (beyond the IEC 61215 standard; ISOS-T-3I) was tested by placing the packaged devices in the thermal cycling test chamber (THERMOTRON) and performing temperature cycling from –40 °C (30 min dwell) to 85 °C (30 min dwell) at a ramp rate of 100 °C h^{-1} . Temperature was measured by the chamber system. The initial average PCE was 23.5% for Me-4PACz- NiO_x devices and 24.0% for D4P- NiO_x devices. The devices were taken out of the chamber after 50, 100, 150, 200, 300 and 500 cycles for the measurements.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available from the corresponding authors on reasonable request. Source data are provided with this paper.

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Author contributions

J.C. and Y.H. conceived the idea and designed the experiments. Y.H. directed and supervised the project. J.C. developed the debonding technique. J.C. and X.W. fabricated the PSCs. T.W. characterized the material chemical and mechanical properties. J.L., H.L., S.L., X.G., R.G., Z.J., Q.Z., Y.W., Z.S., H.Z., M.Z., Z.X. and W.R.L. conducted material and photovoltaic characterizations for the PSCs. H.Y.C. and W.Y. assisted with the mechanical test. S.X. assisted with the XAS characterization. X.Y. performed the molecule design and synthesis. D.L. helped review the manuscript. J.C. and Y.H. analysed data and wrote the manuscript. All authors read and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Solar Cells Reporting Summary

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► Experimental design

Please check: are the following details reported in the manuscript?

1. Dimensions

- Area of the tested solar cells ☒ Yes ☐ No The area of tested solar cells is 0.050 cm².
- Method used to determine the device area ☒ Yes ☐ No Metal aperture mask (black color)

2. Current-voltage characterization

- Current density-voltage (J-V) plots in both forward and backward direction ☒ Yes ☐ No We provide J-V curves for both forward and backward direction in Supplementary Fig. 31.
- Voltage scan conditions ☒ Yes ☐ No See details in Method.
For instance: scan direction, speed, dwell times
- Test environment ☒ Yes ☐ No See details in Method.
For instance: characterization temperature, in air or in glove box
- Protocol for preconditioning of the device before its characterization ☐ Yes ☒ No No preconditioning protocol is used.
- Stability of the J-V characteristic ☒ Yes ☐ No We provided MPPT data in Supplementary Fig. 36.
Verified with time evolution of the maximum power point or with the photocurrent at maximum power point; see [ref. 7](#) for details.

3. Hysteresis or any other unusual behaviour

- Description of the unusual behaviour observed during the characterization ☐ Yes ☒ No Negligible hysteresis was observed.
- Related experimental data ☒ Yes ☐ No Forward and reverse JV scan can be found in Supplementary Fig. 31.

4. Efficiency

- External quantum efficiency (EQE) or incident photons to current efficiency (IPCE) ☒ Yes ☐ No We provide EQE measurement in Supplementary Fig. 31.
- A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator ☒ Yes ☐ No We compare the integrated J_{sc} with one from JV scan. The difference between the integrated J_{sc} from EQE and J_{sc} from JV scan is within 2.2 % difference, which is within accuracy confidence of the measurements. Integrated J_{sc} from EQE is higher than J_{sc} from J-V scan.
- For tandem solar cells, the bias illumination and bias voltage used for each subcell ☐ Yes ☒ No We didn't fabricate tandem solar cell.

5. Calibration

- Light source and reference cell or sensor used for the characterization ☒ Yes ☐ No See details in Method.
- Confirmation that the reference cell was calibrated and certified ☒ Yes ☐ No Our solar simulator was calibrated by Si reference cell. (certificated by Newport.)

Calculation of spectral mismatch between the reference cell and the devices under test

☒ Yes
☐ No

See details in Method.

6. Mask/aperture

Size of the mask/aperture used during testing

☒ Yes
☐ No

An optical aperture mask (0.050 cm²) was used.

Variation of the measured short-circuit current density with the mask/aperture area

☐ Yes
☒ No

No significant variations was observed.

7. Performance certification

Identity of the independent certification laboratory that confirmed the photovoltaic performance

☒ Yes
☐ No

We provide certification data in Supplementary Fig. 32.

A copy of any certificate(s)

Provide in Supplementary Information

☒ Yes
☐ No

We provide certification data in Supplementary Fig. 32.

8. Statistics

Number of solar cells tested

☒ Yes
☐ No

20.

Statistical analysis of the device performance

☒ Yes
☐ No

We provide statistics in Fig. 4f and Supplementary Fig. 33.

9. Long-term stability analysis

Type of analysis, bias conditions and environmental conditions

For instance: illumination type, temperature, atmosphere humidity, encapsulation method, preconditioning temperature

☒ Yes
☐ No

We provided stability data in Fig. 4g and Supplementary Fig. 36.